Study Notes

Economics -Terminology

What is Economics?

We study economics because

- Human wants are unlimited
- Resources are scarce
- Study of Economics helps in allocation of limited resources for meeting human need

Economics is the social science that studies the production, distribution, and consumption of goods and services. Economics focuses on the behaviour and interactions of economic agents and how economies work. It's a study of economic activities, where one can fulfil their unlimited wants through limited resources.

What is Bond?

- Government issues bond to raise money for spending on various development projects like in making schools, dams, roads, hospitals, etc. Likewise private companies raise the debt for expanding business and financing their operations.
- Bonds are also called **fixed income securities** because their returns are predictable as they have a defined interest rates. A Bond
  1. is a financial security which represents the promise of its issuer (normally a company or a government) to repay a loan
  2. has a specified time period and a specified rate of interest
  3. can then be bought and sold to other investors, over and over again
  4. When the rate of interest falls, bond prices rise (and vice versa)
- A bond is a fixed income instrument that represents a loan made by an investor to a borrower (typically corporate or governmental). ... Bonds are used by companies, municipalities, states, and sovereign governments to finance projects and operations, e.g. 91 days Treasury Bill, G-Sec, Corporate Bonds. Owners of bonds are debtholders, or creditors, of the issuer. A bond has a maturity date when the principal (and usually including interest) amount is due to be repaid.
- It is a debt instrument created for the purpose of raising capital. They are essentially loan agreements between the bond issuer and an investor, in which the bond issuer is obligated to pay a specified amount of money at specified future dates.
- The most common types of bonds include municipal bonds and corporate bonds. **Capital**

- Capital is a term for financial assets, such as funds held in deposit accounts, as well as for the physical factors of production; that is, manufacturing equipment.
- While money is used to purchase goods and services for consumption, capital is more durable and is used to generate wealth through investment.
- The four major types of capital include debt, equity, trading, and working capital.

i. ***Debt Captal-*** A business can acquire capital through the assumption of debt. Debt capital can be obtained through private sources, such as friends and family, financial institutions and insurance companies, or through public sources, such as federal loan programs.

ii. ***Equity Capital***-Equity capital is based on investments that, unlike debt capital, do not need to be repaid. This type of capital can include private investment by business owners as well as contributions derived from the sale of stock.

iii. ***Working Capital-***Defined as the difference between a company's current assets and current liabilities, working capital measures a company's short-term liquidity—more specifically, its ability to cover its debts, accounts payable, and other obligations that are due within a year. In a sense, working capital is a snapshot of a firm's financial health.

iv. ***Trading Capital-***Trading capital refers to the amount of money allotted to buying and selling various securities. Generally, trading capital is distinct from investment capital because it is reserved for more speculative ventures. Trading capital is sometimes referred to as "bankroll‖.

Commodity

- Anything that is bought and sold for money is a commodity, including produced goods and services, inputs (such as capital or raw materials), and even labour.
- A commodity is a basic good used in commerce that is interchangeable with other commodities of the same type. Commodities are most often used as inputs in the production of other goods or services. ... When they are traded on an exchange, commodities must also meet specified minimum standards, also known as a basis grade.

Corporation/Company

- independent legal entity and association of people
- separate from the individuals who own it
- provides for limited liability
- well-suited to the joint-stock form of ownership
- A large company or group of companies authorized to act as a single entity and recognized as such in law.

Credit

- The ability of a customer to obtain goods or services before payment, based on the trust that payment will be made in the future.
- Ability to purchase something without immediately paying for it – through a credit card, a bank loan, a mortgage, others.

Debt

- Debt is an amount of money borrowed by one party from another. ... A debt arrangement gives the borrowing party permission to borrow money under the condition that it is to be paid back at a later date, usually with interest.

Depreciation

- Reduction in the value of an asset over time, due in particular to wear and tear. **Economic growth**
- An increase in the amount of goods and services produced per head of the population over a period of time.

Economies of scale

- Economies of scale are the cost advantages that enterprises obtain due to their scale of operation, with cost per unit of output decreasing with increasing scale.

Factors of production

- Factors of production are the inputs used in the production of goods and services. For example land, labor, capital and entrepreneurship. Knowledge has also been considered as a potential new factor of production.

Gini Coefficient

- The **Gini index** or **Gini coefficient** is a measurement tool developed by the Italian statistician Corrado,**Gini** in 1912. It is used for measuring economic inequality or income distribution among the residents of a nation.
- A value of zero Gini coefficient means perfect equality, i.e. an economy in which everyone has the same income. A Gini coefficient of 1 means maximal inequality or where only one person has all the income.

- Inflation

- Inflation is increase in the price level of goods and services in an economy over a period of time.With price rise, each unit of currency buys fewer goods and services; therefore inflation reflects a reduction in the purchasing power of the money.

Globalisation

- The process by which businesses or other organizations develop international influence or start operating on an international scale.

Equity

- A stock or any other security representing an ownership interest, which might be in a private company in which case it's called private equity.

Goods

- Goods are tangible products which are produced in the economy. There are four types of goods – Public goods, private goods, common resources and club goods.

Macro-economics and Micro-Economics

- ***Macroeconomics*** is the study of economics concerned with large-scale or general economic factors, such as interest rates and national productivity.
- ***Micro-economics*** is the study of economics concerned with single factors and the effects of individual decisions.at an individual, household or company level.

Speculation

- Speculation is the purchase of an asset with the hope that it will become more valuable in the near future.

Tariff

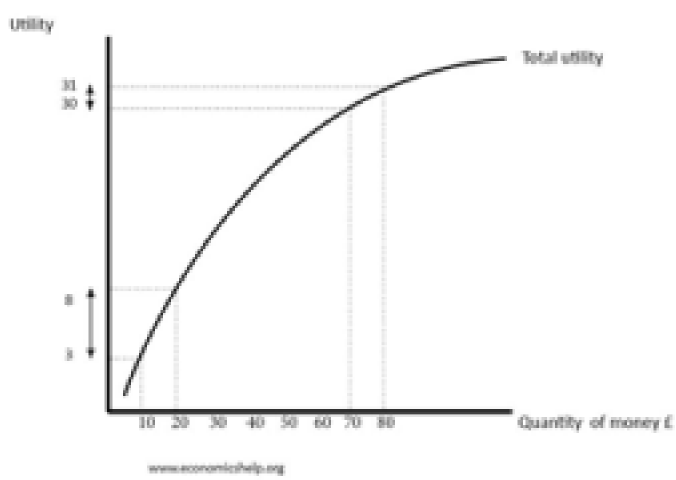
- Is a tax levied on imported goods and services.
- Tariff is imposed to stimulate the domestic production of the goods or services. **Recession and Depression**
- In economics recession is negative economic growth (measured by real GDP) for two consecutive quarters or six months. In this unemployment increases and government has to do more borrowings. Recessions can occur due to wars, famine, floods or market fall. There can be phases of recession due to business cycles.
- A depression is a severe and prolonged downturn in economic activity. In economics, a depression is commonly defined as an extreme recession that lasts two or more years. A depression is characterized by economic factors such as substantial increases in unemployment, a drop in available credit, diminishing output, bankruptcies and sovereign debt defaults, reduced trade and commerce, and sustained volatility in currency values. In times of depression, consumer confidence and investments decrease, causing the economy to shut down.

Cartel

- A group of independent producers coming together for increasing their collective profits by price fixing means or limiting supply
- Cartels generally are there to keep the selling price higher, however some cartels are there to control the prices of purchased inputs
- Governments and antitrust laws attempt to deter or forbid cartels

Diminishing Marginal Utility (DMU)

- The Law of Diminishing Marginal Utility states that all else equal as consumption increases the marginal utility derived from each additional unit declines. Marginal utility is derived as the change in utility as an additional unit is consumed. Utility is an economic term used to represent satisfaction or happiness.



Primary Products

- Primary products are goods that are available from cultivating raw materials without a manufacturing process. Significant primary product industries include agriculture, fishing, mining, and forestry.

Economic Development

- Economic development' is a process in which a nation is being improved in the sector of the economic, political, and social well-being of its people**.**

Economic Growth

- Economic growth is a quantifiable concept, which means it can be measured. We have three methods (Income method, Expenditure method and Production/Output method) and various aggregators like GDP , GNP , NDP , NNP , GVA etc to measure the economic growth.
- ―Economic growth means increase in economic activities production/output/income/expenditure) during a given period of time (month, quarter or year) for a country (or state/company/city etc).‖
- Economic growth is essential for the well-being of any society as it leads to jobs creation, increased income, wealth creation etc.

Measurement of Growth

- Economists developed various tools to measure the economic growth , for example , GDP GNP GVA NNP NDP etc. These are called ―Macroeconomic aggregators‖ or ―Macroeconomic identities‖.
- Given below are few macroeconomic indicators used by IMF, World Bank today. · ―Output‖ - It is the aggregated monetary value of all the final goods and services produced in an economy during a given period of time.
- ―Consumption goods‖- These include goods like food, clothing, services that are consumed by the end users. These are also called consumer goods.
- ―Capital goods‖- These are also final goods but not ultimately consumed. Instead these aid and enable the production process of further goods and services. For example- tools, implements, machineries etc. They gradually undergo wear and tear.

- —Depreciation‖ – The amount used in accounting for the regular wear and tear of capital (goods) is called depreciation. The government decides the rate of depreciation for different assets and hence it varies from economy to economy and asset to asset.
- —Nominal rate‖- Nominal rate refers to the interest/growth/exchange rate before taking inflation into account.
- —Real rate‖- A real rate is an interest/growth/exchange rate that has been adjusted to remove the effects of inflation to reflect the real income or growth etc.
- For example, if we are getting 8% interest on our Fixed deposits in Banks and the inflation rate is 4% in the economy then even if the nominal rate is 8% the real rate (and return) is only 4%.
- Same applies to economic growth rate also. If we say the economy grew at 10% without accounting for the inflation, the growth might be due to the inflated monetary values of goods and services produced without any real increase in the production. Hence we always use Real growth rate.
- —Base year‖- It is the year used as the reference year for comparing any ratio, rate or index. For example the base year for GDP and IIP has been revised to 2011-12 by the government to reflect the correct picture of economy. This means 2011-12 would be the reference year for comparing or calculating these identities.
- It reflects the changes in purchasing power.
- —At factor cost‖- When we calculate any economic aggregator like GDP , GVA etc —At factor cost‖ , it includes the total monetary value paid to all the factors of production(which in essence is the cost/price of the production) but does not include the taxes or subsidies involved in final market prices.
- —At Basic prices‖- Basic prices include the total monetary value/cost of production AND taxes and subsidies which are Production specific and not PRODUCT specific.
- GDP at Basic prices = GDP at factor cost + (Production tax-Production subsidies) · —At market prices‖- These includes monetary value of production plus Production tax less Production subsidies plus Product tax less Product subsidies.
- GDP at market price = GDP at Factor cost + (Production tax-Production subsidy)+(Product taxes-Product subsidy)

Macroeconomic identities

1. Gross Domestic Product (GDP)

- This is the most commonly used aggregator to measure economic growth.This is calculated through Production / Product / Output / Value added method.
- ―GDP is the total monetary value(hence the term Gross) of all the final goods and services produced(hence the term Product) within the boundary of a nation(hence the term Domestic) during given time period which is usually a year or quarter‖.
- Annual percentage change in GDP is the growth rate of economy.
- GDP at market prices= Gross value added at factor cost + (Production taxes Production subsidies) + (Product taxes-Product subsidies)
- We need to keep in mind whether the aggregators are compared at current or constant prices.
- GDP at Constant prices are calculated for the base year we use in calculation(for GDP calculation Central Statistics Office uses 2011-12 as the base year) while GDP at Current prices are calculated for current year. The difference is of inflation adjustment.
- In India National Income Accounting is done by Central Statistics Office. It shifted the base year and also changed the method of calculation National Income from January 2015. The base year is changed to 2011-12 from 2004-05.
- New series also introduced the concept of Gross Value Added (GVA) too reflect the qualitative aspect of economy growth.

2. Net Domestic Product (NDP)

- ―NDP is arrived after deducting depreciation (consumption of fixed capital) from GDP‖
- NDP at market prices= GDP- Depreciation
- This is used for various macroeconomic analysis. Also it reflects the achievements of the economy in Research and development sector which contributes in cutting down the depreciation value (wear and tear of capital goods)

3. Gross National Product (GNP)

- This is also calculated through Production / Product / Output / Value added method, however there is one difference between GDP and GNP.
- —GNP is the total monetary value of all the goods and services produced by all the members of the same NATIONALITY (whether within the boundary of the nation or outside) during a given period of time, which is usually a year or quarter‖.
- To calculate GNP from GDP we add —Net Income from Abroad‖
- GNP = GDP + (Net Income from Abroad)
- Net Income from abroad= Income of Indian nationals outside Indian territory Income of foreign nationals within Indian territory
- Under this —Income from abroad‖ term we include - Trade Balance Interest of External Loans, Private Remittances etc.
- Based on the level of openness of economy, immigration and emigration status of the country etc , —Net Income from abroad‖ can be positive or negative. Accordingly GNP can be more or less than GDP.
- GNP = GDP - Income from Abroad
- Few important points regarding GNP is listed below
- It is both a QUANTITATIVE as well as QUALITATIVE measure of economic growth because it not only takes internal but external strength of economy too under calculation
- It is not preferred by countries to act as Official measure of Economic Growth because of its dynamic and interdependent nature. In today's globalised world —Net Income from abroad‖ is a very vulnerable term and can inflate or deflate the economic growth measurement by a huge margin. Hence ,countries prefer to use GDP over GNP.
- However, IMF uses GNP as the —National Income‖ of the member nations and ranks them accordingly.

4. Net National Product(NNP)

- NNP is calculated after adjusting depreciation from GNP.
- NNP = GNP – Depreciation
- Dividing NNP by the total population gives us Per Capita Income of the country which is income per head per year.
- Per Capita Income = NNP / Population
- In India, this is categorised as the ―National Income‖ of the country.

5. Gross National income (GNI)

- GNI is the aggregated primary incomes of all the institutional units in the economy. This is calculated through Income method.
- Primary income includes compensation of employees , property income , Taxes as primary income of governments etc.
- It can be calculated from GDP by adding Net Primary income to GDP. However , Net Primary income here menas Net Primary Income from Rest of the World. o GNI = GDP + Net Primary Income from Rest of the World
- If we see the formula of GNP , we will find GNI and GNP as same. This is true too.
- One thing we need to understand here that whether we measure economic growth by Income method , Production method or Consumption-Expenditure method , the value remains the same. Let us take an example to understand the concept.
- Suppose the economy has only one Production unit utilising all the available factors of production. In such case the cost of production for the production unit will essentially be the income for all the factors of production and again the same will be the expenditure of consumers for these products. Hence the value remains the same.

6. Net National Income (NNI)

- NNI is calculated from GNI after adjusting for the Depreciation.
- NNI = GNI – Depreciation
- This is also a measure through Income method.

7. Gross Value Added (GVA)

- GVA as a concept was introduced in India in 2015 when CSO revised the National Income Accounting with its New series Data and Methodology. o This is also essentially a Production/Product/Value added method measure however here we measure not only the cost of production BUT the Gross Value Added during the production process. This gives us a better quality picture of strength and advancement of the economy.
- To understand the concept of Value added method let us take an example. o Suppose a readymade garment factory produces shirts. The raw material for these shirts would be textile , thread , buttons etc. Let us take the value/cost of the raw materials used as Rs.100. After producing shirts from this textile it would be sold for Rs.250. In this case we say the value added during the production of these shirts is Rs.150(Rs.250-Rs.100). However the raw materials we used here were also products of some other raw materials (textile made of yarn) and during that production also some value addition happened which brought the cost of these raw materials as Rs.100.
- Hence when we add the gross value addition of the entire production cycle here , it comes as Rs. 150 + Rs. 100 = Rs. 250. That is the cost of production too , butcalculating the same through value addition method gives us a detailed picture of how much value addition took place at what stage of production. This analysis helps in understanding which sector is progressing and which needs better support to improve the economy.
- Under New Series of National Income Accounting , GVA at Factor cost and GVA at Basic Prices are calculated as intermediary steps to calculate GDP at market prices.
- GVA at factor cost = Gross Value added(Output) - Intermediate Consumption **-** (Production Taxes -Production Subsidies)
- GVA at basic prices = Gross Value added(Output) - Intermediate Consumption **+** (Production Taxes -Production Subsidies)
- GDP at market prices = GVA at basic prices +(Product Taxes -Product Subsidies)

Important Economic indicators

1. Index of Industrial Production

- This is not a measure to capture the growth of whole economy, however this is one of the important indicators to analyse and assess the progress of economy. This Index gives a picture of Industrial activities in the economy comprising all the important sectors and items.
- It is published monthly by CSO. The Base year for calculating the same has been revised to 2011-12 in 2015 (earlier it was 2004-05) . This captures the progress made under three sectors (Manufacturing , Mining and Electricity) and total 865 items.

Sector	Share under IIP
Manufacturing	77.63%
Mining	14.37%
Electricity	7.99%

- It includes all the 8 core industries of the economy. The share of these core industries (in decreasing order) is listed as below.

Industry	Share under IIP
Refinery Products	28.04%
Electricity	19.85%
Steel	17.92%

Industry	Share under IIP
Coal	10.33%
Crude Oil	8.98%
Natural Gas	6.88%
Cement	5.37%
Fertilizers	2.63%
Total	**40.27%**

2. Purchasing Managers' Index

- This is an important indicator for the progress of both manufacturing as well as services sector(Note that IIP only includes manufacturing industries and not services).
- This index is prepared based on a survey which includes perception of business people about certain business variables compared with the last month. It is calculated separately for manufacturing and services sector and also a combined index is released.
- A value less than 50, indicates contraction in the sectors involved and a value more than 50 indicates expansion in economic activities.
- It is published monthly by Japanese firm Nikkei but compiled and constructed by Markit Economics in India.
- Difference between PMI and IIP

	PMI	IIP
Published By	Nikkei	CSO
Frequency	Start of the month	End of the month
Sectors covered	Both Manufacturing and Services	Only Manufacturing
Bae year	No Base year	2011-12
Database	Survey of 500 companies	Annual Survey of Industries

New Series of National Income Accounting by CSO

- CSO introduced some changes in January 2015 to the National Income Accounting. The growth data released using these changes are termed as New Series Data. · Important changes introduced are listed below
- The Base year was changed from 2004-05 to 2011-12. It was last changed in 2010.
- CSO did away with GDP at factor cost calculation and adopted international practice of calculating industry wise estimates of gross value added GVA at basic prices. Apart from this, Net Value Added and GDP at market prices are calculated under new series.